SMARTBRIDGE

SKILL WALLET
A SmartBridge Product

Date	02 July 2026
Team ID	
Project Name	**Credit Card Approval Prediction**
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	**Yes**	Used 4-space indentation consistently across all Python scripts and notebooks.
2	Proper File Structure	Files and folders are logically organized	**Yes**	Project organized into Data, Notebook, Model, and Output directories.
3	Meaningful Variable Names	Variables reflect their purpose clearly	**Yes**	Variables such as `application_df`, `credit_df`, `final_df`, `X_train`, `y_test` clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	**Yes**	Functions like `create_target()`, `evaluate_model()` improve readability.
5	Code Comments	Inline and block comments explain logic	**Yes**	Important preprocessing, feature engineering, and model evaluation steps are documented with comments.
6	Modular Design	Code is split into reusable functions/modules	**Yes**	Separate reusable functions for target creation, evaluation, preprocessing, and visualization.

7	No Redundant Code	Duplicate or unused code is removed	**Yes**	Duplicate code minimized by using reusable functions and loops.
8	Error Handling	Exceptions and errors are handled gracefully	**Partial**	Basic error handling implemented; advanced exception handling can be added for production deployment.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Loading Module	Python (Pandas)	Dataset Import	High
2	Data Cleaning Module	Python (Pandas, NumPy)	Missing Value Handling	High
3	Target Creation Function	Python	Target Variable Generation	High
4	Model Evaluation Function	Scikit-learn	Used for all ML Models	High
5	Visualization Module	Matplotlib, Seaborn	EDA & Result Visualization	High

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	**5**	Well-organized project structure with logical workflow from data loading to model evaluation.
Readability	**5**	Meaningful variable names, proper formatting, and sufficient comments improve readability.
Reusability	**5**	Functions and preprocessing steps are modular and reusable for similar classification problems.
Documentation / Comments	**4**	Adequate comments are provided throughout the notebook; additional function docstrings can further improve documentation.

Overall Score	**4.75 / 5**	The project follows good coding practices, modular design, and machine learning workflow standards.